PaliGemma · arXiv 2407.07726 · Sep 16

Stakeholder

seat 4 · 18+3

Open 3B substrate → robot foundation models

Paper club · VLA Architectures · dark academic

Why this paper is on the calendar

Commodity open 3B VLM → robot foundation-model substrate

Open base (not chat) ~3B VLM for transfer

SigLIP-So400m eyes + Gemma-2B language → default init for π_0 / $\pi_{0.5}$ / OpenPI
BEHAVIOR’25 1st place stack = SigLIP → PaliGemma → Action Expert (~26% q)

Perception / grounding — NOT a classical planner

Buy decision

Init from
paligemma-pt-{224,448}
not custom glue

Budget

Stage3-like FT
hours on modest
TPU/GPU per domain

Multi-cam

Often stay @224;
escalate res only for
OCR / fine grasp

Architecture · open base ~3B VLM

SigLIP-So400m + linear connector + Gemma-2B · Prefix-LM

Vision

SigLIP-So400m
(shape-opt ViT)

Stage0 contrastive
Public checkpoint
224 / 448 / 896

→

Connector

Linear projection
(zero-init)

MLP: no gain (§5.5)
Dense visual tokens
→ LLM embedding

→

Language

Gemma-2B v1.0
pretrained (not IT)

Prefix-LM masking
full attn: img+prefix
causal: suffix

Token layout: [image…] BOS prefix… SEP(\n) suffix… EOS PAD…

loc tokens: 1024 bins · seg tokens: 128 VQVAE codes · skills via task prefixes (no GPT-4 IT)

Perception / grounding backbone — NOT a classical planner or policy

Training stages 0 → 3

Unimodal priors → multimodal skills → resolution → per-task transfer

Stage 0	Stage 1	Stage 2	Stage 3
Unimodal	**Multimodal**	**Resolution**	**Transfer**
SigLIP + Gemma public ckpts off-the-shelf	~1B ex @224 nothing frozen ~350B tokens	+50M @448 +10M @896 OCR/det upweight	Per-task FT lr=1e-5 default 20min–10h TPU

Key Stage1 choice: do NOT freeze ViT (slow LR warmup) — freezing LM hurts a lot

Prefer released paligemma-pt-{224,448} as substrate for π_0 / OpenPI / BEHAVIOR forks

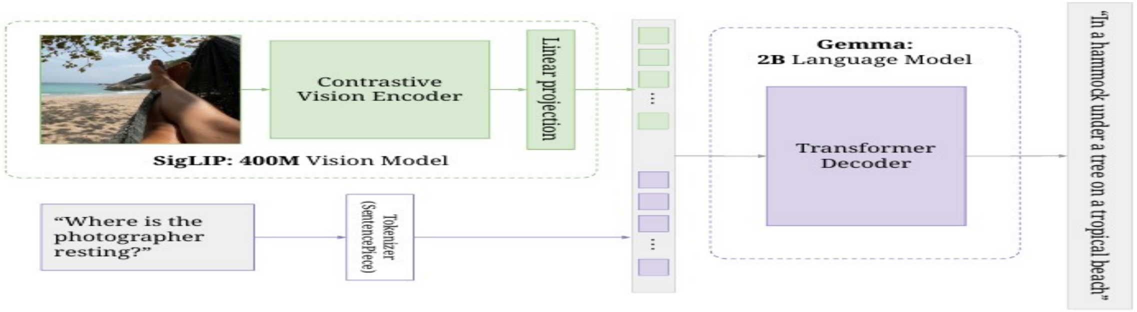

Figure 1 | PaliGemma's architecture: a SigLIP image encoder feeds into a Gemma decoder LM.

These were then further scaled up by, among others, Flamingo [6], BLIP-2 [62] and, PaLI [23]. Finally, most recent works [7, 70, 87, 113] perform an additional "instruction tuning" step that is intended to make the raw model more user-friendly. In addition to building systems, several recent more systematic studies [59, 81, 107] aim to find out what really matters in VLMs. PaliGemma is an open base VLM without instruction tuning, and this report answers a few more questions regarding what matters. More discussion in Appendix A.

3. Model

In this section we present details about PaliGemma's architecture and training. Several of our decisions are further ablated in Section 5.

At a high level, PaliGemma is a VLM, taking as input one or more images, and a textual description of the task (the prompt or question, which we often refer to as the `prefix`). PaliGemma then autoregressively generates a prediction in the form of a text string (the answer, which we often refer to as the `suffix`).

This simple image+text in, text out API is flexible enough to cover many standard tasks, such as image classification, captioning, visual question-answering and dialogue. Additionally, as shown in the literature, by converting more complex structured outputs into "text", this API can also cover more tasks such as: detection [22], instance segmentation [25, 115], panoptic segmentation, depth prediction, colorization, and many more [56, 73, 139]. This conversion can be hand-engineered and task-specific, such as done in pix2seq [22] for detection, or learned as is the case for segmentation [56] and dense output tasks in general.

During PaliGemma's pretraining, we limit ourselves to "text" covering natural language, object detection, and instance segmentation, but this API remains versatile and the pretrained models can be finetuned for other output types.

3.1. Architecture

PaliGemma consists of three components:

- An image encoder, for which we use a publicly available SigLIP [133] checkpoint, specifically the "shape optimized" [5] ViT-So400m image encoder. This model was contrastively pretrained at large scale via the sigmoid loss, and has shown state-of-the-art performance, especially for its small size.
- A decoder-only language model, for which we use the publicly available Gemma-2B v1.0 [82] raw pretrained checkpoint, which strikes a great balance between performance

Paper architecture / method context

Headline transfers · Table 1 selected

Resolution is load-bearing for OCR/doc — DocVQA is the poster child

Task	224	448	896
COCOcap	141.9	144.6	—
VQAv2	83.2	85.6	—
ScienceQA	95.4	95.9	—
DocVQA ANLS	43.7	78.0	**84.8**
TextVQA	55.5	73.2	**76.5**
RefCOCO A	75.7	77.9	**78.7**

DocVQA 43.7 → 84.8 with resolution · prefer pt-448 when OCR/widgets matter

Club chain · SigLIP → PaliGemma → Action Expert

Why an open 3B base VLM is load-bearing for this club’s robot FM work

SigLIP So400m
Eyes
contrastive

→

PaliGemma
Base 3B VLM
Prefix-LM

→

π_0
VLA init
from PG

→

$\pi_{0.5}$ / OpenPI
Open robot
stacks

→

BEHAVIOR ’25 1st
Action Expert
$q \approx 0.26$

PaliGemma = the critical open bridge

Closed VLMs cannot seed OpenPI forks. Loc/seg + OCR@448+ = grounding API.
Failures often sit in action expert / stages / data — but these are the eyes.

Risks & buy guidance

Private Stage1 data

Cannot fully retrain from scratch at paper scale; treat ckpt as prior

License / gating

Gemma + SigLIP + HF gating; check terms before commercial forks

Base ≠ chat

Not instruction-tuned; expect Stage3-like FT for UX / robots

Prefer paligemma-pt-{224, 448}

224 for multi-cam / latency · 448 when OCR / fine spatial matters · 896 sparingly
Budget Stage3-like FT; unfreeze vision early (slow warmup); don’t expect chat out of the box

Stakeholder takeaway

Start from open paligemma-pt-{224,448}

Budget Stage3-like FT · multi-cam often @224
Escalate resolution for OCR / fine spatial
Own the risks: private data · license · base≠chat

Differentiate downstream — not by reinventing the 3B glue

Hand off: Reviewer owns critique · Empiricist owns LoRA/res ·
Archaeologist owns lineage · Visionary owns 2026 adaptation

Club thread · perception vs planner

PaliGemma IS

- Open base VLM for transfer
- Loc/seg + OCR grounding API
- Default π_0 / OpenPI eyes+language
- Prefix-LM observe+goal shaping

PaliGemma is NOT

- A classical motion planner
- A closed-loop policy
- An instruction-tuned chat bot
- Proof of BEHAVIOR wins alone

Hand-offs

Empiricist: LoRA / freeze / res ladder pressure-tests.
Visionary: adaptation (action expert, stages, data) > backbone swap in 2026.

Training stages 0 → 3

Unimodal priors → multimodal skills → resolution → per-task transfer

Stage 0
Unimodal

SigLIP + Gemma
public ckpts
off-the-shelf

→

Stage 1
Multimodal

~1B ex @224
nothing frozen
~350B tokens

→

Stage 2
Resolution

+50M @448
+10M @896
OCR/det upweight

→

Stage 3
Transfer

Per-task FT
lr=1e-5 default
20min–10h TPU

Key Stage1 choice: do NOT freeze ViT (slow LR warmup) — freezing LM hurts a lot

Prefer released paligemma-pt-{224,448} as substrate for π_0 / OpenPI / BEHAVIOR forks

Stakeholder close

Start from paligemma-pt-{224,448} · own the risks · hand off deep dives